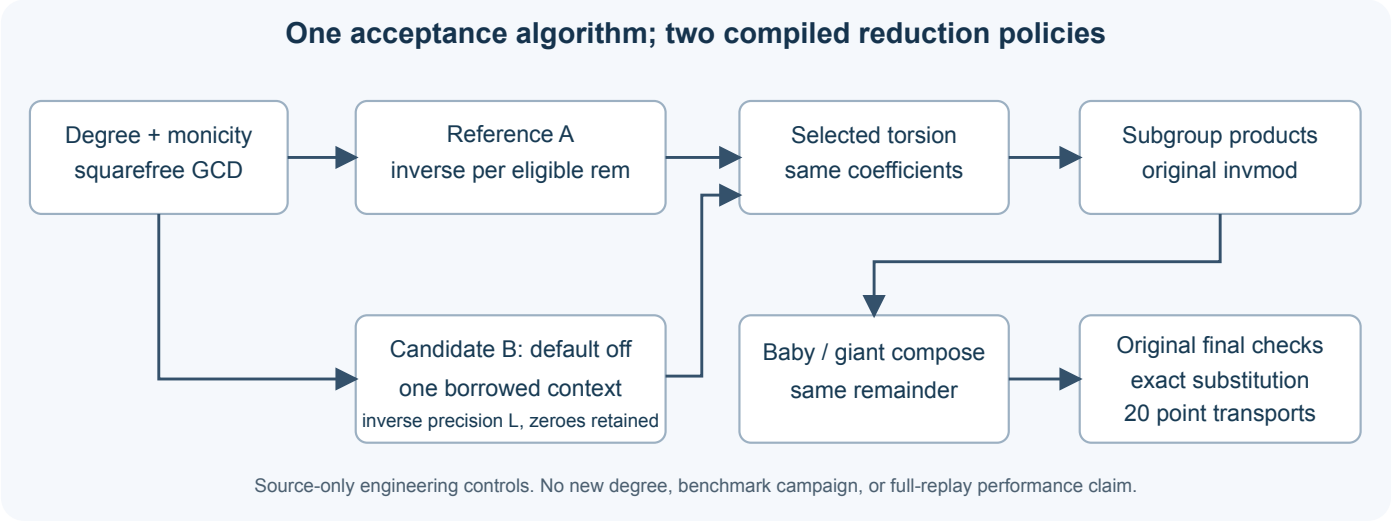
Clean native source/control checkpoint

The largest-degree goal remains active. This checkpoint preserves exact default/reference behavior and validates the default-off cached kernel-reciprocal implementation. It establishes no new admitted degree or performance result.



Evidence gate	Observed outcome
Clean build and installed byte identity	Both policies dirty=false; compiled bytes equal installed bytes
Complete native suite	128 passed, 0 failed per arm; one parent-invoked child helper ignored
Complete mathematical output	Six byte-identical JSON pairs; zero exclusions; all 20 transports per map
Rehashed/adversarial negatives	Nine trace pairs plus two actual-cache negatives agree
Lost/zero-RSS controls	16 retained fail-closed injection receipts per arm; two worker shapes
Deadline controls	Two retained timeout receipts per arm
Exact-source Lean	Two checked lemmas; standard propext/Quot.sound only
Remote custody / independent review	LOCAL ONLY / NOT STARTED; source gate incomplete

Exact source: 015fa7bb504ef63953e0e04e464bb266f4f207bf. Both builds use rustc 1.93.1, the same lockfile, target and release profile; only the one explicit feature differs. The source and version JSON identify the policy separately from mathematical certificates.

Executable SHA256. Reference: `2ff7d57174cda2d0e90088eb6ce217cbd1826e7697ed12ec7770d3d8f395d379` Candidate: `305bdff7db7bc25a2ccdce3f44e71c3d47269e76afa0ecd1703803365ab9b207`

Native manifest and all control hashes: native-manifest.json. Native archive: 1,261 regular files,2,049,977 bytes; every archived hash replayed. Executables and target directories are excluded; the compiled Lean object is included. The archive and exact receipt are beside this report.

Root failures, retakes and limits

The clean default-parallel root library run failed 4146/2, with 117 ignored. Both failures were unchanged JV solver admission-count tests (21 and 22 admitted cases versus thresholds 25 and 30). The complete JV group then passed 14/0/1 ignored serially. The entire root library then passed 4148/0/117 ignored with one test thread, without skipping a test or changing any threshold or source.

The same root test executable was used by the earlier default-parallel pass, the later default-parallel failure and the full serial pass. Its SHA256 is `e6b0366e26794fe703741ba309cd3c7cb168150c827fa2a23ba068038aec7399` Both the failed result and successful retakes are retained in root-gates/. This is observed variation; its cause is not established. Parallel stability is not asserted.

Accounting. The complete serial library body reported 1047.40s. Its OS receipt recorded 1047.87s real, 799.80s user and 101.98s system. The intervals overlap. Native builds/tests retain their own OS receipts. These are engineering costs on a shared Apple M4 Pro host, not isolated performance observations. Earlier development CPU costs remain incomplete.

Input-preparation control. Native generation used the accepted P192/13 map only, produced eight dataset cases, and returned PASS with zero unavailable RSS samples for both compiled variants. The two datasets are byte-identical: `2b5bb3f2cdd04ddb3ab068e99c7a81e22e3801bcfd0471759ca993ec9e5fd871` Generation computes reference expectations in both variants. These bounded controls do not measure cached speed. The large datasets remain locally retained and hash-bound in generation-control.json.

Proof scope. IDC1h41e9973f839c85bf is a finite-field evaluation supplement. Lean checks the scalar integer identity and an integer coefficient-convolution precision lemma. Full formal-series uniqueness and quotient-reversal formalization remain open; no fresh lean4checker was available.

Failures preserved. The initial native alias failure, initial Lean lemma-name failure, later metadata path error, root parallel failure and all corrected controls remain accessible. The first native archive was sealed before the last full-root retake; that final retake is a separate immutable addition.

Next gate. Remote custody awaits authorization, and independent source/resource review has not started. TASK-de577e is not complete. No measure/analyze campaign, 50341 extraction/replay, larger construction, compact target, default activation or board promotion ran. Source-only PR 1991 remains the implementation PR. All 54 standard prime-field boards and the separate reviewed M44497 compact result are unchanged.